Case Report

Analysis of the Safety Factor and Mechanical Properties of a Weight-Reduced Gearbox using a Simulation Study

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ABSTRACT

The design of engineering components always includes two parts: component design and material selection. Additionally, gearboxes are key components in the power transmission system; therefore, great care must be taken in their design. In the present study, an attempt has been made to present a new design for advanced gearboxes. Two ferrous and non-ferrous alloys were selected for gear and component materials, and mechanical analyses were performed on them. SolidWorks software was used for the design. The results showed that the selected steel and titanium alloys are close competitors in terms of mechanical properties and safety factors, and both materials can be used in specific design conditions. However, due to the lighter weight of titanium compared to steel, designers are advised to conduct further studies in the field of developing components for various industries with non-ferrous alloys, as they play an effective role in reducing fuel consumption, conserving energy, and continuous improvement in various industries and products.

Introduction

Approximately 23% of global energy consumption occurs due to tribological contacts [1]. Therefore, efficiency plays a significant role in mechanical and industrial applications, and in the field of machinery technology, and one of the primary goals is to achieve maximum efficiency. To attain this objective, it is crucial to make appropriate component selections, ensure the compatibility of component pairs, and establish suitable operating conditions, subsequently operating mechanisms to follow these conditions, as this can result in substantial gains. The transmission of power from a source, such as an engine or motor, through a machine to an output actuation is one of the most common machine tasks. The design of a system to transmit power requires attention to the design and selection of individual components (gears, bearings, shafts, etc.) [1]. It becomes crucial in this situation to comprehend the classification and working principles of power transmission systems. Mechanical transmission can be categorized into friction and engagement drives based on the principle of transmission [2]. On the one hand, in

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a friction drive, power and motion are transferred via friction, such as in a belt drive. Although friction transmission cannot be used for high-power occasions, overload slippage plays a role in buffering and protecting the transmission device. On the other hand, the engagement drive relies on the interaction of the active and passive parts or intermediate parts to transfer power and motion, such as in gear and chain drives. The engagement drive can be used for high-power occasions. Although it has a good transmission accuracy, it requires high manufacturing and installation tolerances [3].

Due to their extensive use and performance benefits, gear transmissions are especially important among engagement drives. The development of gear transmission is moving towards high-load, high-temperature, and high-speed applications at present, which demand higher requirements for their life and reliability [4]. Mechanical gear drives exhibit important advantages such as high efficiency, accurate gear ratio, mechanical robustness, and relative ease of manufacture. Additionally, these are ideal for low, medium, and high power transmission. Gears use rotation to transfer torque and alter speeds, delivering power and motion from a prime mover to a driven machine or load [5]. The design of the gear drive system needs to consider the backlash between gears. If the backlash is too small, it will affect transmission efficiency; however, if the backlash is too large, it will affect transmission accuracy, while the tooth surface impact will produce vibration and noise, thereby affecting gear life [6]. The most important component of a mechanical power transmission system is the so-called gearbox, primarily due to the role it plays in the system. The gearbox can transmit mechanical power as a speed multiplier or speed reducer. Typically, a gearbox comprises a gear train inside a housing with an input shaft and one or more output shafts. The gear train is a multiple set of gears, meshed together to create movement [5]. The main design criteria for gearboxes are: (i) efficiency and heat balance, (ii) power density and load capacity, and (iii) noise, vibration, and harshness. The underlying physical principles can influence each other, so a compromise is required based on the design aims. The product lifecycle analysis superimposes the main design criteria [7].

Understanding and optimizing the efficiency of a gear mechanism has paramount importance in various industrial applications. Gearbox efficiency is defined as the ratio between input power and output power. In an ideal situation, the input power should be the same as the output power, that is, the efficiency should be 100%. However, in reality, gearbox efficiency can often be 90% or less due to energy losses and friction. The efficiency of the gearbox depends on many factors. The following factors play a significant role in determining the efficiency of a gearbox. Gear type, gear ratio, gearbox size, materials and manufacturing geometry, lubrication, load conditions, operating conditions, and operating temperature can be listed as some of those factors [8].

Reducing the weight of gearboxes has become a central focus in modern mechanical design, driven by the growing demand for energy efficiency, high performance, and environmental sustainability. Lighter gearboxes help decrease the overall mass and rotational inertia of mechanical systems, resulting in faster dynamic response, reduced energy consumption, and improved motion control benefits that are especially critical in applications such as electric vehicles, aerospace systems, and industrial automation. Several strategies are employed to achieve weight reduction, including geometric redesign, material substitution, and the adoption of advanced manufacturing processes. Structural techniques such as hollowing, pocketing, and topology optimization are commonly used to remove excess material from gear components and housings while maintaining mechanical strength and stiffness. At the same time, replacing conventional materials such as steel or cast iron with lighter alternatives, including aluminium alloys, magnesium, polymer composites, or hybrid material configurations, can significantly reduce mass without compromising durability [9-14].

This study aims to present a thorough analysis of mechanical power transmission systems, emphasizing gearbox design. Efficiency, structural optimization, and weight reduction are prioritized, and future development is

guided by both theoretical underpinnings and real-world case studies.

Materials and Methods

In the design and manufacturing of gears, it is essential to select materials that exhibit high resistance to both corrosion and bending stresses. In this study, with the aim of lightweight gear design, several commonly used engineering alloys were initially considered as potential candidates. After a series of experimental evaluations and comparative mechanical tests, AISI 4340 steel was selected due to its superior overall performance. Notably, AISI 4340 is also the material recommended by by American Gear Manufacturers Association (AGMA) for high-performance gear applications, making it a reliable and well-established choice in the field. Table 1 shows the chemical composition of this alloy.

Table 1 Chemical composition of AISI 4340 steel

Element	C	Mn	Cr	Ni	Mo	Si	Fe
% Wt	0.40	0.73	0.8	1.85	0.28	0.2	Base

In the present study, to simulate the characteristics of the gearbox, in addition to the introduced steel, Ti-6Al-4V alloy was selected. This alloy was selected due to its higher strength-to-weight ratio than steel, to play the role of lightening, which is one of the main objectives of the present study. Table 2 shows the chemical composition of Ti-6Al-4V alloy.

Table 2 Chemical composition of Ti-6Al-4V

Element	Al	V	Cr	Ni	Mo	O	Ti
% Wt	6.21	4.2	0.01	0.05	0.05	0.05	Base

In this study, an actual industrial gearbox was selected as the test sample, and all calculations and experiments were performed on it. An actual industrial gearbox was selected as the test sample, and all calculations and experiments were performed on it. Figures 1-5 show the 3D model and 2D drawing of the parts investigated in this study. All design was done by authors of this article.

To examine the properties, various parameters including the safety factor were calculated using mathematical equations and the results were compared. Analysis software was used to analyse the results. In addition to the above, to measure the performance of the component under difficult conditions, the component was placed in a simulator (SolidWorks Simulation environment was used) at a temperature of +250 degrees Kelvin (-23 degrees Celsius) to examine its characteristics. This temperature was used to evaluate the design characteristics. At this temperature, a stress of 500 MPa was applied to the surface of the sample to enable analysis of the strength of the gear teeth. This reduction in temperature helps to determine the design factors and performance of the part at low temperatures.

Results and Discussion

Theoretical Calculation

Engineers recognize that reducing the weight of a system by 43.56%—while maintaining volume, safety margins, mechanical strength, and the ability to withstand both static and dynamic loads, as well as fatigue—represents a crucial advancement. Such a reduction can significantly occur in decrease energy losses, lower fuel consumption, and improve transport efficiency and ease of handling. With these advantages in mind, the following analysis has been carried out to examine the relevant parameters in detail.

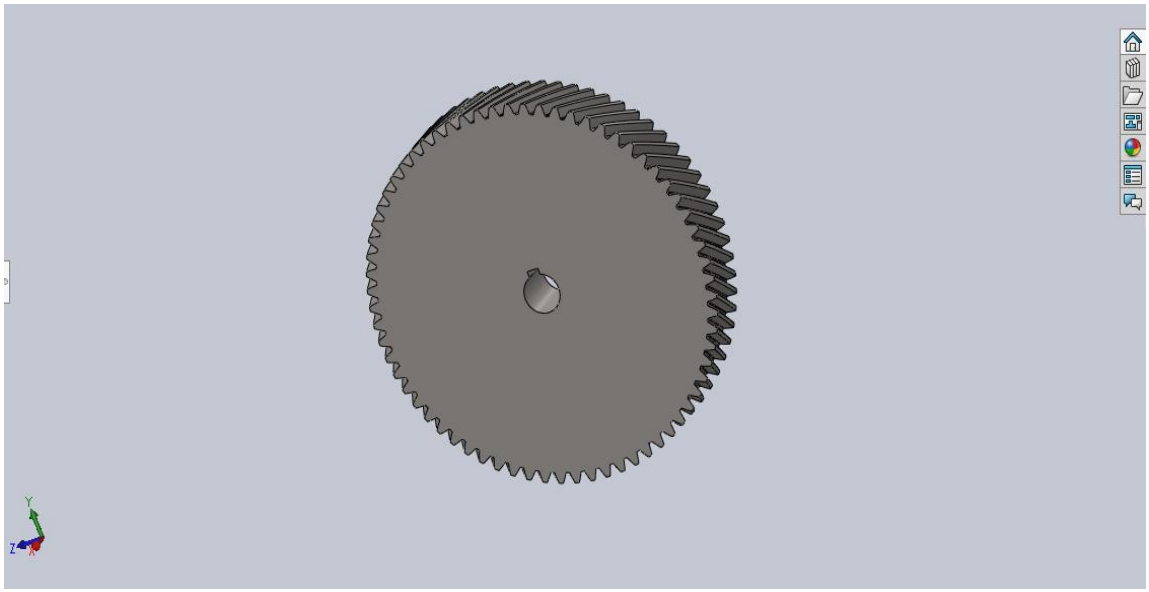


Figure 1 3D model of gear designed by authors

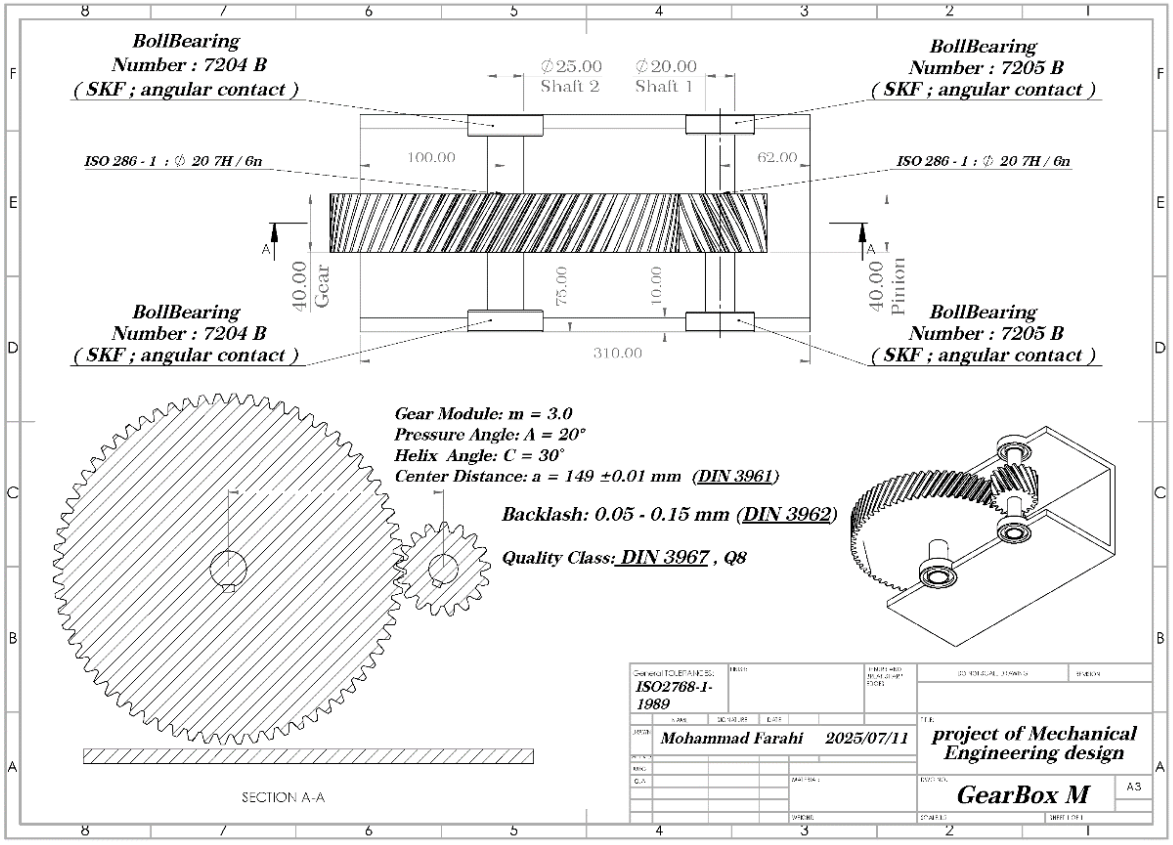


Figure 2 2D drawing of gearbox designed by authors

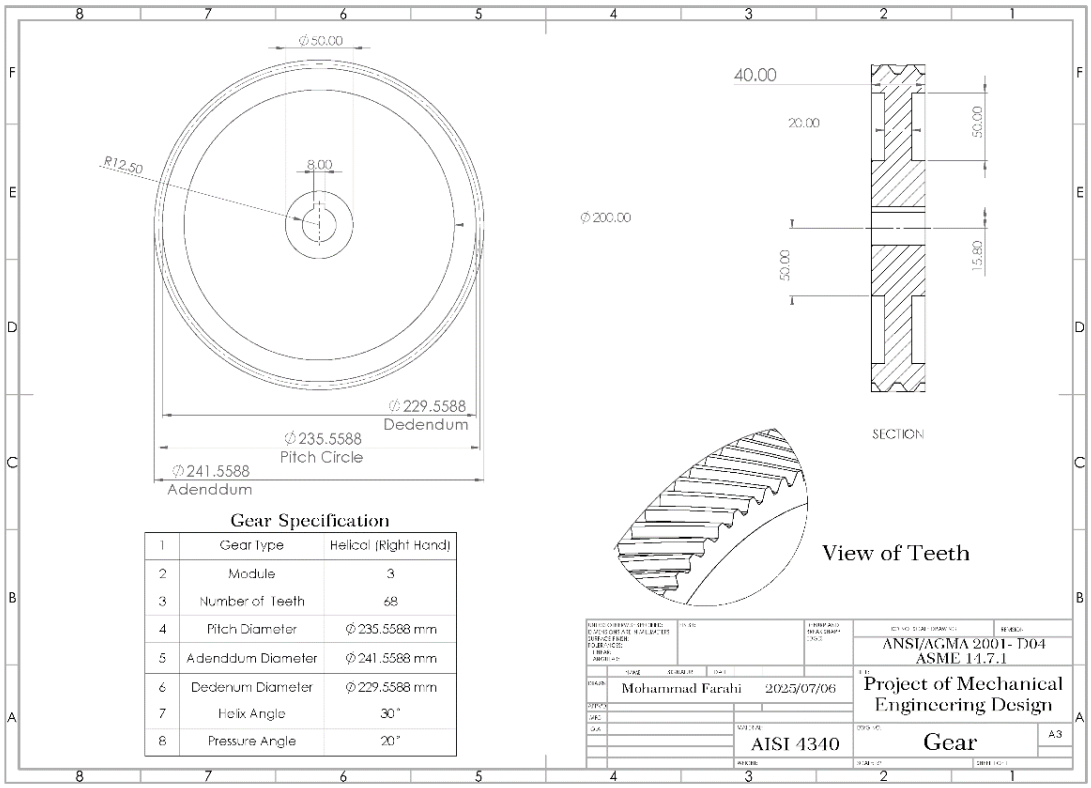


Figure 3 2D drawing of gear designed by authors

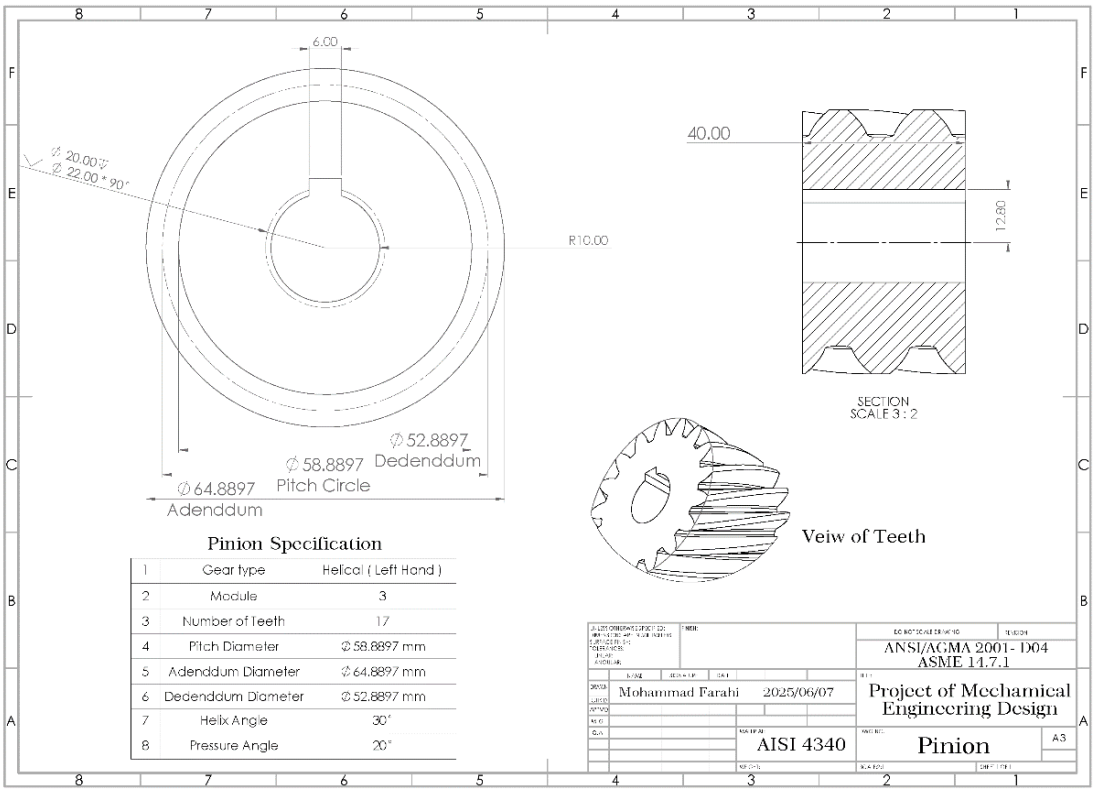


Figure 4 2D drawing of Pinion designed by authors

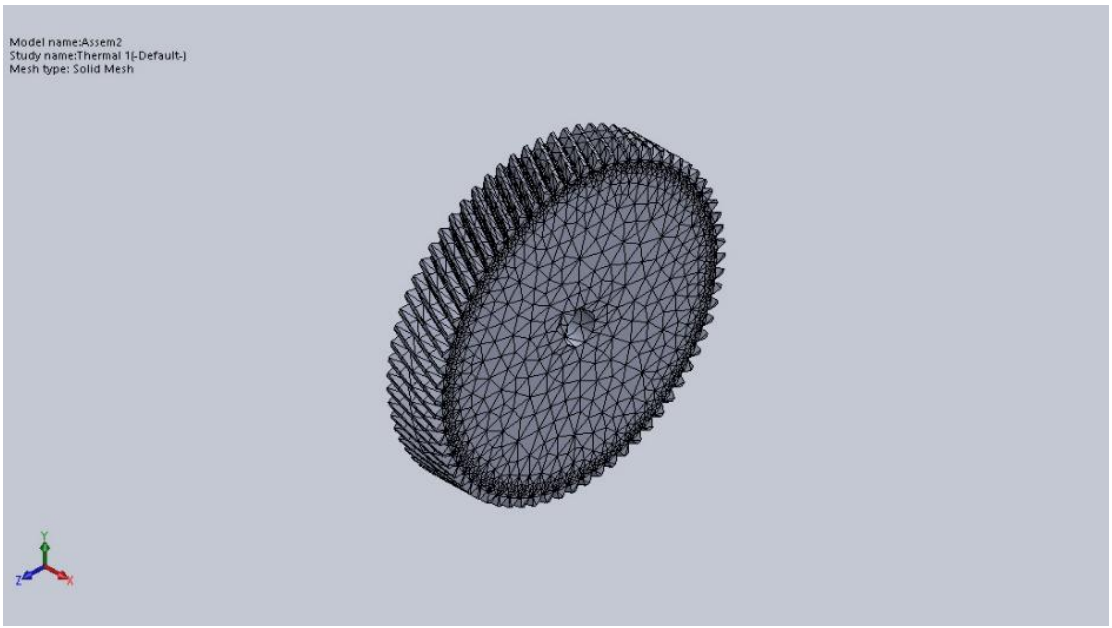


Figure 5 Sample meshing, mesh type: solid mesh, max mesh size: 0.2 inch

Table 3 AISI 4340 and Ti-6Al-4V physical properties

Material	Density (g/cm^3)
AISI 4340	7.85
Ti-6Al-4V	4.43
Difference	3.42
Weight Reduction in this gearbox	43.56%

Ti-6Al-4V is approximately 43.56% lighter than AISI 4340, while offering comparable mechanical properties. With proper design and analysis, it can serve as a viable alternative in many structural applications.

When digging into the gearbox analysis, following the guidelines from groups such as AGMA, the American Society for Metals, and the American Institute of Aerospace Association, results show that two factors are very important in the gearbox system.

1. Safety factors related to bending stress.
2. Safety factors related to contact stress.

Pinion Bending Stress

For the pinion's bending stress, the Ti-6Al-4V material showed a safety factor of 3.8192, while the AISI 4340 steel came in at 4.5370. This means titanium alloy's safety factor was just 0.7178 lower than the steel's. Now, whether this difference matters depends on the design's minimum safety factor. It might be negligible, but if both materials need to have the same safety factor, some of the affected components must be slightly reinforced.

Gear Bending Stress

The results were similar for the gear's bending stress. The Ti-6Al-4V shows a safety factor of 4.1079, while the AISI 4340 steel was 4.8799. Again, the Ti-6Al-4V alloy was only 0.772 lower than the steel. Just like with the pinion, this difference could be small enough to ignore depending on the proposed design's baseline

safety factor. However, if perfect parity is needed, a slight reinforcement of the affected parts could be in this regard.

Pinion Contact Stress

As can be seen in Figure 6, for the pinion's contact stress, Ti-6Al-4V boasted a safety factor of 1.5994, while AISI 4340 was at 1.154210. This clearly shows that Ti-6Al-4V alloy performed better than steel in this topic.

Gear Contact Stress

The trend continued with the gear's contact stress. Ti-6Al-4V came in at 3.1989, and AISI 4340 was 2.3084. Once again, the proposed titanium material outperformed the steel, resulting in an impressive 0.8905 increase in the safety factor.

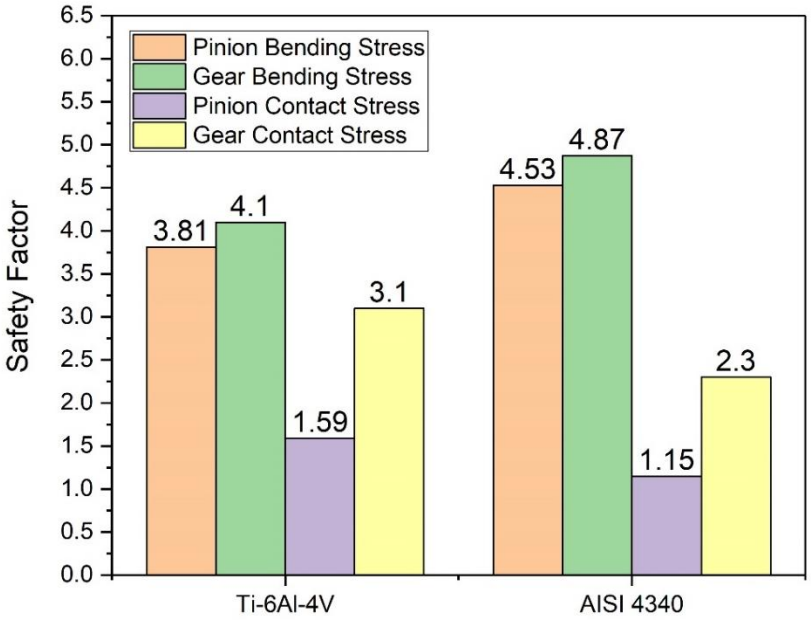


Figure 6 Ti-6Al-4V alloy VS AISI 4340 safety factor values

Concerning all these findings, it is clear that switching to Ti-6Al-4V titanium alloy instead of common steel materials such as AISI 4340 (which AGMA often recommends for gears) could reduce the weight of the gearbox or gear power transmission components by a significant 43.56%. Furthermore, given that it either maintains or even improves safety factors, volume, mechanical properties, dynamic and static load capacity, fatigue resistance, and other critical aspects, we recommend making this material change.

Simulation Results

Theoretical calculations may always differ from reality; therefore, it is necessary to evaluate the conditions of the part in the simulator. To achieve a rigorous simulation, the analysis was performed at a temperature lower than the ambient temperature. Figure 7 shows the results of the displacement after applying stress to one surface of the gear. As it is clear, the displacement caused by applying surface stress to one side of the gear in Ti-6Al-4V alloy is greater than that of steel. It is noteworthy that most of the displacement occurs is in the teeth, which may cause problems in the operation of the gearbox.

The reason for this and its importance can be found in the reduction of temperature. As the theoretical results showed, there was no significant difference between steel and titanium alloy in the design and safety factors. This is because in general design cases, temperature does not interfere. However, by

reducing the temperature to sub-zero levels, structural changes led to further damage to the material and the strength of titanium decreased. This illustrates the importance of the design temperature and the consideration of environmental parameters.

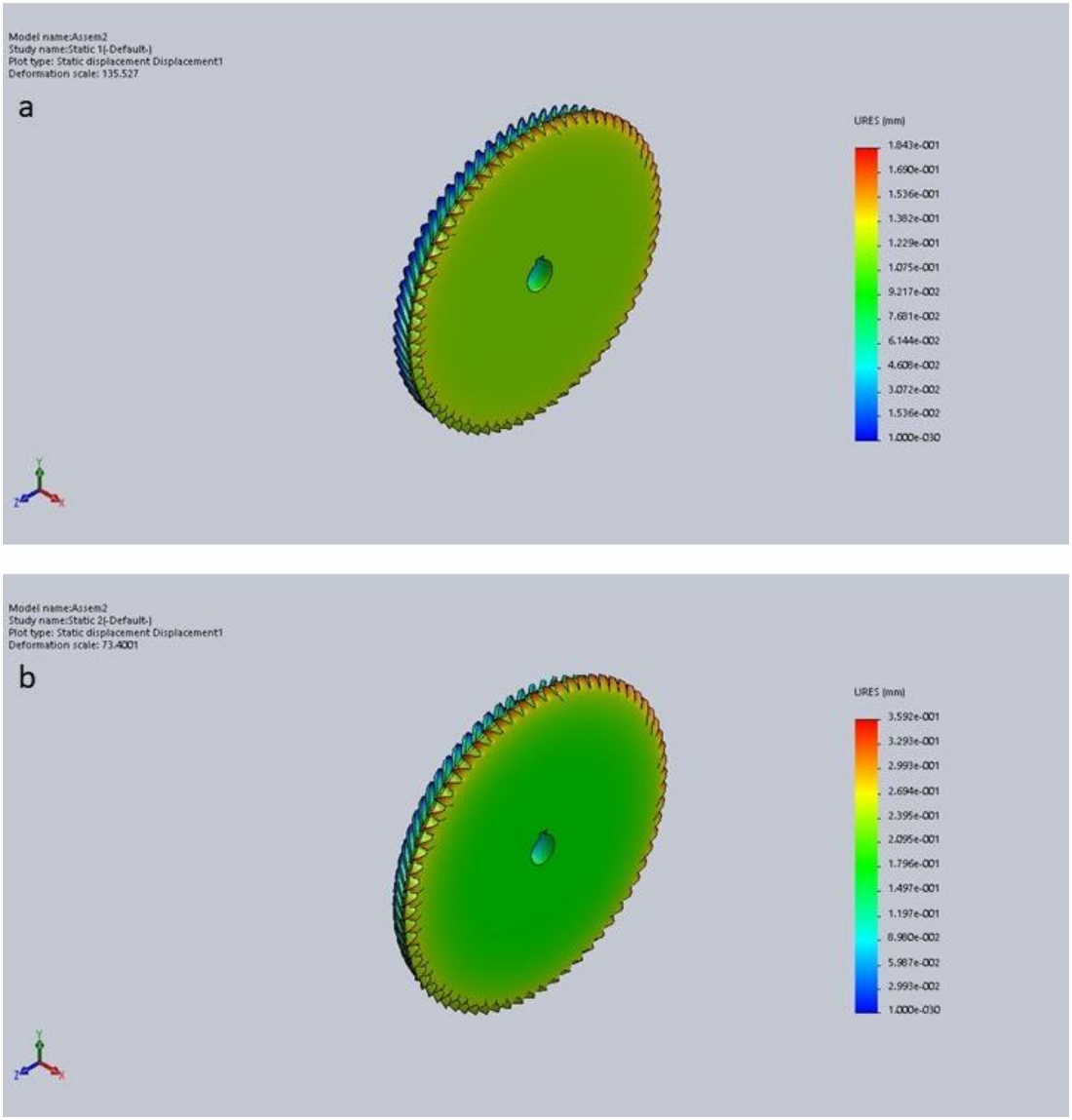


Figure 7 Displacement results for alloys, a) AISI 4340 and b) Ti-6Al-4V

Figure 8 also shows the values of the von Mises stress, which according to the results have surpassed the yield point in sections under multiaxial loading. This indicates that both parts may not have sufficient strength under the aforementioned stress and temperature conditions, where steel exhibits better performance; however, it can still be investigated further. It is emphasized again that in this study, strict conditions were considered for the design of applied stress and temperature so that possible differences could be observed.

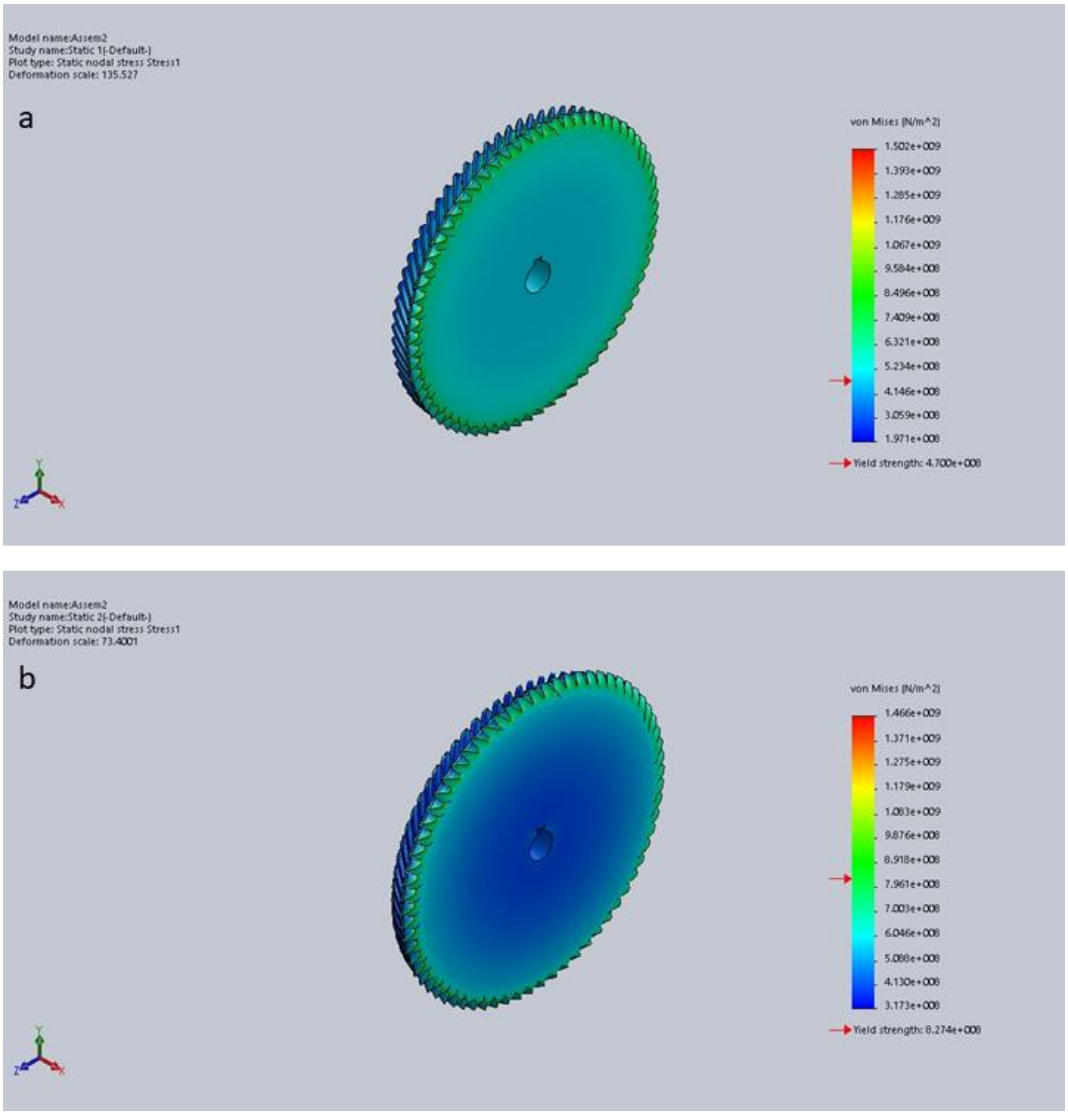


Figure 8 The von Mises stress results for alloys, a) AISI 4340 and b) Ti-6Al-4V

Conclusion

In general, the results of the present study showed: (1) In the designed gearbox, a new structure was used that can improve the performance of the gearbox. It is suggested that researchers focus more on new structures in the future. (2) The materials tested included two groups of ferrous and non-ferrous. The results showed that ferrous materials can be useful in functions that require greater mechanical strength. (3) Despite the acceptable performance of ferrous materials in the design, non-ferrous materials and titanium alloys also showed good resistance and, in addition, their weight was reduced by about 50% in the design. (4) This gearbox design can lead to a reduction in fuel consumption in the future, which is one of the main constraints globally. Therefore, it is necessary to use this design to ensure the performance of the gearbox by controlling the coefficients. (5) Reducing the weight of this gearbox design can be applied in special designs; therefore, according to the results obtained, it is suggested that researchers focus more on non-ferrous alloys in the design of advanced parts in the future. (6) It is recommended that in the future, an operational prototype be built according to the presented drawings so that its results can be compared with current research.

Conflict of Interest

The authors declared no conflict of interest.

Data Availability Statement

All data are available upon request.

Authors' Contributions

O. Ashkani: Project manager, conseptization, final writing, and final approval. M. Farahi: Theoretical calculation, 2D and 3D design, gearbox designer, first draft, and final approval. A. Azizkhani and A. Ghahremani: 2D and 3D design, software simulation, first draft, and final approval.

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